

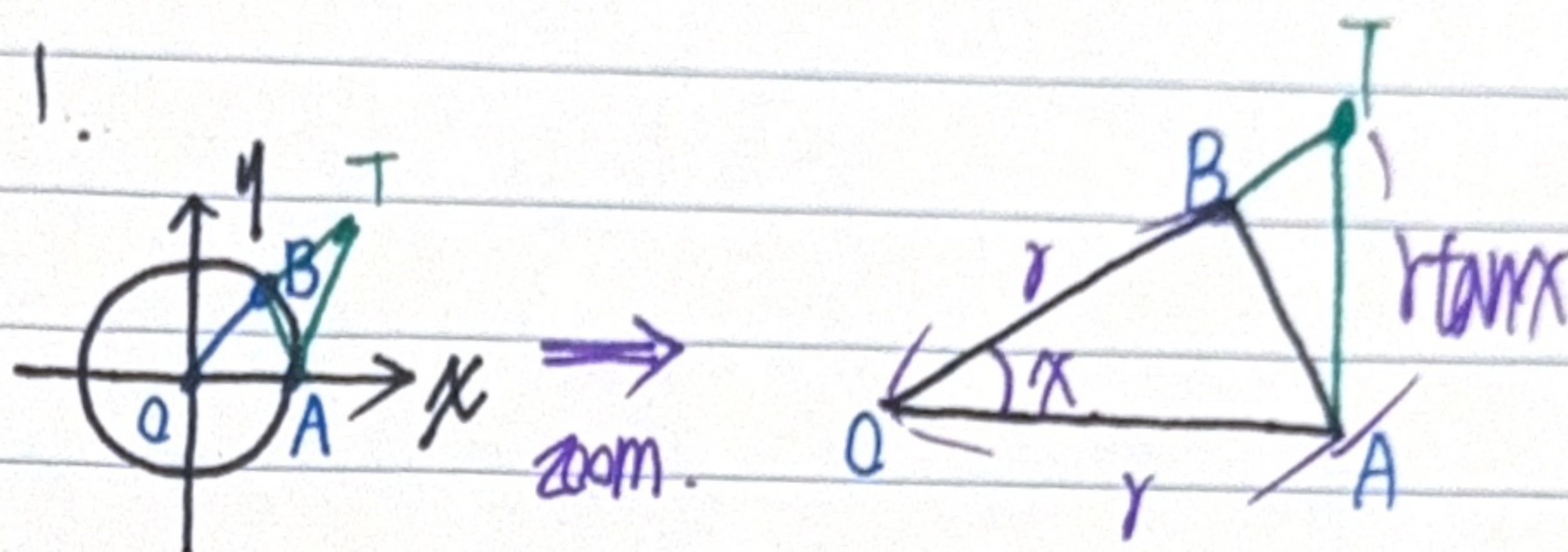
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Def.

when x unit is rad.

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1.$$

P.f.



when $0 < x < \frac{\pi}{2}$

$$\triangle OAB < \triangle OAT.$$

$$= \frac{1}{2}r^2 \sin x < \frac{1}{2}r^2 x < \frac{1}{2}r^2 \tan x.$$

$$\therefore \sin x < x < \tan x.$$

$$= 1 < \frac{x}{\sin x} < \frac{\tan x}{\sin x} \left(\Rightarrow \frac{1}{\cos x} \right) \Rightarrow 1 > \frac{\sin x}{x} > \cos x.$$

when $x \rightarrow 0^+$

$\cos x \rightarrow 1$

when $-\frac{\pi}{2} < x < 0$

let x be $-\theta$.

$$\begin{aligned} \lim_{x \rightarrow 0^-} \frac{\sin x}{x} &= \lim_{\theta \rightarrow 0^+} \frac{\sin(-\theta)}{-\theta} \\ &= \lim_{\theta \rightarrow 0^+} \frac{\sin \theta}{\theta} = 1. \end{aligned}$$

$$\therefore \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1.$$

$$\Rightarrow \lim_{x \rightarrow 0^+} 1 > \lim_{x \rightarrow 0^+} \frac{\sin x}{x} > \lim_{x \rightarrow 0^+} \cos x.$$

$$\therefore \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$